

Conditional Cross-Lingual Steering: Closing the Multilingual Reasoning Gap in Small Language Models

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Abstract

Small Language Models (SLM) are becoming increasingly important for on-device deployment, but their English-centric pretraining creates a measurable performance gap on multilingual reasoning tasks. While activation steering has been applied to properties such as tone and language style, its potential to close cross-lingual reasoning gaps in SLMs remains unexplored. We propose a cross-lingual activation steering approach using contrastive pairs drawn from questions answered correctly in English but incorrectly in a target non-English language, and evaluate it with and without conditional activation steering across SLMs from the Llama-3.2 and Qwen-2.5 families. We find that conditional steering consistently outperforms vanilla steering across models and languages. Our results demonstrate that targeted activation steering can close reasoning gaps between English and other languages in SLMs without finetuning. We release our experimental codebase and reproducibility artifacts at [this GitHub repository](#).

1 Introduction

The concept of developing efficient Small Language Models (SLM) has become increasingly important over the past few years due to the need to have portable and environmentally friendly versions of models that still retain good performance (Liu et al., 2024; Wang et al., 2025a). In particular, running an efficient off-cloud model on a phone is important, as on-cloud API usage tends to incur large computational costs and privacy concerns (Wang et al., 2025a). One might even push this further and explore pushing the boundaries of SLMs for multilingual tasks (with the corresponding application being the development of portable devices with strong native multilingual capabilities). Since most language models have

an English-centric pretraining corpus (Wang et al., 2025b), there tends to be a gap between the performance of English and non-English languages (Li et al., 2024). We confirmed this disparity by running our own benchmarking across several models, the results of which are detailed in [Appendix A](#). While increasing model size generally improves performance across all languages (Kaplan et al., 2020), it is still important to investigate methods to improve multilingual performance for smaller, fixed sizes for contexts in which we might only be able to deploy SLMs.

An emergent finetuning-free method of improving model performance is to add steering vectors to a model’s hidden layer representations. The foundational assumption for steering vectors is the linear representation hypothesis, which states that high-level concepts are represented linearly as directions in some representation space (Park et al., 2024). For instance, one might posit that there exists a direction within a model’s hidden representation that encodes the concept of ‘rudeness’. Compared to finetuning, this approach is computationally cheaper (Ghosh et al., 2025).

The primary research question investigated in this report is as follows: **Can cross-lingual activation steering reduce the performance gap between English and non-English languages for SLMs on reasoning tasks?** Specifically, our investigation is limited to language models with up to 3 billion (3B) parameters. We propose a novel selection of contrastive pairs to extract steering vectors to bridge the gap between a model’s non-English and English reasoning abilities when applied at test time. Subsequently, we evaluate this method over 8 (non-English) languages using varying sizes of Llama-3.2 and Qwen-2.5 architectures. Leveraging an implementation of conditional steering (Lee et al., 2025), we demonstrate that the performance disparities can be mitigated without the need for finetuning.

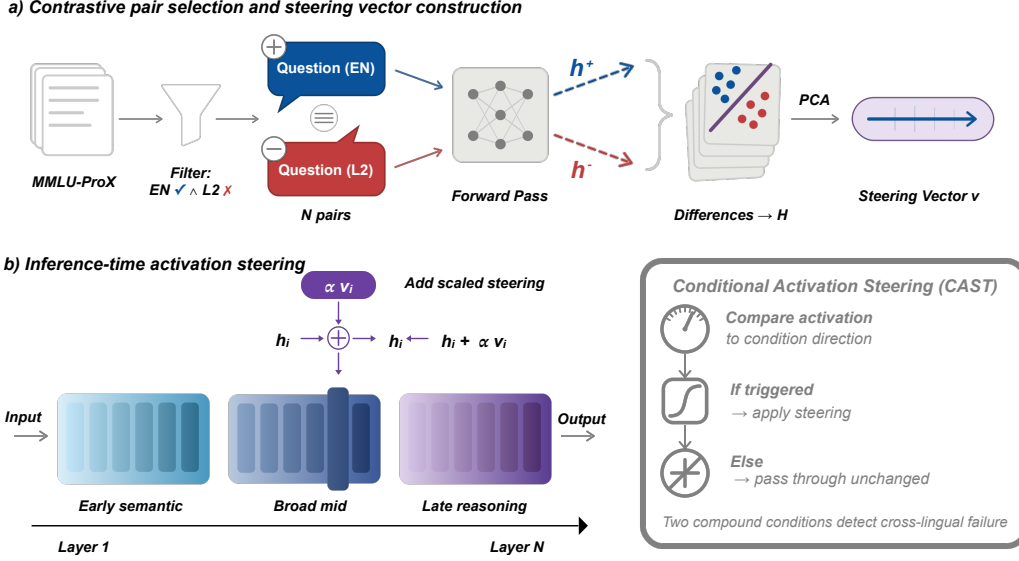


Figure 1: Overview of our methodology. (a) Steering-vector construction from contrasting English (positive) and target-language (L2) (negative) prompts selected under the MMLU-ProX filter. (Section 3). (b) Inference-time activation steering across layer groups, with optional CAST gating.

2 Related Work

2.1 Multilingual Performance in SLMs

The performance gap between English and non-English languages in SLMs widens at smaller parameter counts, consistent with the scaling laws of Kaplan et al. (2020). A key question is whether this gap can be addressed without retraining. Schut et al. (2025) provides evidence that multilingual language models internally route non-English inputs through English-language representations. If such a shared representational pathway exists, it suggests that an English-derived signal in activation space may be transferable to non-English inputs. This is the assumption that underlies our cross-lingual steering method.

2.2 Activation Steering

Subramani et al. (2022) showed that latent vectors extracted from language models can be used to steer their outputs. Panickssery et al. (2024) later demonstrated that averaging the difference in activations between contrastive prompt pairs produces a direction that reliably steers model behaviour. Compared to finetuning, activation steering is computationally lightweight and avoids unintended side effects of gradient-based parameter updates (Ghosh et al., 2025).

Subsequent work has applied activation steering across tone and style (Turner et al., 2025),

instruction-following (Stolfo et al., 2025) and safety refusals (Ghosh et al., 2025; Lee et al., 2025). A complementary direction is test-time optimisation of steering vectors, which selects the vector that minimises output entropy rather than relying on a fixed contrastive extraction procedure (Kang et al., 2026). In this work, we extend activation steering to the cross-lingual reasoning setting in SLMs.

2.3 Extracting Steering Vectors

The original approach of Subramani et al. (2022) optimises the vector directly via gradient descent, fixing its norm and maximising the likelihood of a target output. Panickssery et al. (2024) instead computes the mean difference of activations across contrastive pairs, which is simpler and does not require backpropagation. Lee et al. (2025) refines this further by taking the first principal component of the matrix of mean-centred differences, which is more robust to noise in the contrastive pairs and is the aggregation method we adopt in this work.

2.4 Conditional Activation Steering (CAST)

Applying a fixed steering vector indiscriminately to all inputs potentially degrades performance on inputs that do not require correction. Lee et al. (2025) addresses this with CAST, which will be elaborated on in Section 3.2.

CAST was used to selectively refuse prompts while preserving general model utility (Lee et al., 2025). We adapt CAST to the cross-lingual setting, using it to detect whether the model is likely to fail on a non-English input before applying the steering vector.

2.5 Multilingual Benchmarks

The original Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al., 2021) established the standard for multi-task language understanding evaluation using 4-option multiple-choice questions. Wang et al. (2024) extended this to MMLU-Pro, which increases the option set to 10 and replaces knowledge recall questions with reasoning-focused items. MMLU-ProX (Xuan et al., 2025) further extends MMLU-Pro to 29 languages across 14 subject categories, making it the primary benchmark in this work.

For cross-dataset transfer experiments, we additionally evaluate on Global-MMLU (Singh et al., 2025), a parallel multilingual benchmark covering 42 languages with improved translation quality. We also evaluate on Belebele (Bandarkar et al., 2024), a reading comprehension benchmark. As Belebele shares no domain alignment with MMLU-ProX, it allows us to test the generalisability of our steering vectors beyond their training distribution.

3 Methods

Through benchmarking multiple frontier SLMs on MMLU-ProX across a range of languages (Appendix A), we empirically demonstrated the performance gap between English and non-English languages on these models. In this section, we describe our method using activation steering with the aim to reduce this gap. To implement activation steering, we utilise the code library developed by (Lee et al., 2025).

3.1 Steering Vector Extraction

Similarly to (Turner et al., 2025), the steering vectors in (Lee et al., 2025) are generated by aggregating the hidden activations generated by applying the model to contrasting prompt pairs. A key difference is that (Lee et al., 2025) use Principal Component Analysis (PCA) for aggregation instead of simple averaging as in (Turner et al., 2025). In our case, given the hidden activations of the positive and negative prompts at layer l , de-

noted $\{(h_{l,i}^+, h_{l,i}^-)\}_{i=1}^N$ for a collection of N contrasting pairs, we construct the matrix

$$H = \begin{pmatrix} (h_{l,1}^+ - \mu_{l,1})^T \\ (h_{l,1}^- - \mu_{l,1})^T \\ \vdots \\ (h_{l,N}^+ - \mu_{l,N})^T \\ (h_{l,N}^- - \mu_{l,N})^T \end{pmatrix},$$

where $\mu_{l,i} = \frac{1}{2}(h_{l,i}^+ + h_{l,i}^-)$ for each $i = 1, \dots, N$. The steering vector for layer l , denoted v_l , is given by the first principal component of the matrix H . At inference time, for any input x , the steering vector v_l is scaled by a steering coefficient α and added to the hidden activations at layer l (Turner et al., 2025; Lee et al., 2025), as illustrated in Figure 1(b). In practice, we apply steering simultaneously across layer groups rather than a single layer; the specific layer groups are detailed in Section 4.

3.2 CAST

To mitigate the potential negative effects of indiscriminate steering, we apply steering conditionally. Under the CAST framework (Lee et al., 2025), steering uses a *condition vector* c , applying the update:

$$h_l(x) \leftarrow h_l(x) + \alpha f\left(\frac{h_l(x)^T \pi_c(h_l(x))}{\|h_l(x)\| \|\pi_c(h_l(x))\|}\right) v_l,$$

where $\pi_c(u) = \frac{cc^T}{\|c\|^2}u$ is the projection of u onto the span of c , f is a binary thresholding function and v_l is the steering vector as above. This means that CAST applies activation steering only if the hidden activation $h_l(x)$ is sufficiently well-aligned with c , as measured by the value of f on the cosine-similarity between the hidden activation and its projection onto c . The vector c is generated using contrasting prompt pairs in the same way as the steering vector. As explained in (Lee et al., 2025), it is also possible to use multiple conditions by simply modifying the function f to include multiple conditions in the threshold.

3.3 Contrasting Prompt Pairs

Our objective was to reduce the gap in performance between English and non-English languages by utilising the superior English performance of the models. We hypothesised that comparing how a model processes a question it answers correctly in English versus how it fails on

the exact same question in a target language will isolate the specific activations responsible for this performance disparity. We filtered the MMLU-ProX dataset for questions where the unsteered model succeeds in English but fails in the non-English language. From this filtered subset, we extracted a fixed number of questions to form our contrastive pairs: the English version serves as the positive prompt, while its non-English counterpart serves as the negative prompt.

For the CAST experiments, we use two condition vectors generated from contrasting prompt pairs. For constructing the contrasting pairs, the positive examples for both condition vectors consist of questions in English that the unsteered model answers correctly in English but incorrectly in the target non-English language. The motivation for the negative examples is to describe the cases where we do not want steering to be applied. Therefore, for the first condition vector, the negative examples are given by questions (in the target non-English language) that the unsteered model answers incorrectly in English, and for the second condition vector, the negative examples are given by the questions (in the target non-English language) the unsteered model answers correctly in both English and the target non-English language.

We fixed this method of generating contrasting prompt pairs for all of our experiments and evaluated it comprehensively, as detailed in Section 4, to investigate whether this approach reduces the performance gap.

4 Experiments

For our experimental setup, we evaluate across 2 model families, 5 model sizes, 8 target languages (spanning high and low-resource languages), 3 layer groups, and 16 steering strengths.

Models We evaluate two model families at multiple scales, all with at most 3B parameters: **Llama-3.2** (1B-Instruct, 3B-Instruct) and **Qwen-2.5** (0.5B-Instruct, 1.5B-Instruct, 3B-Instruct). Llama-3.2 was trained predominantly on English data (as mentioned [here](#)), making it a strong candidate to benefit from cross-lingual steering. Qwen-2.5 was trained on a more multilingual corpus (Qwen et al., 2025), providing a contrasting setting.

Datasets Our primary benchmark is MMLU-ProX (Xuan et al., 2025), introduced in Sec-

tion 2.5. For cross-dataset transfer experiments (Baselines 4 and 5), we additionally evaluate on Global-MMLU (Singh et al., 2025) and Belebele (Bandarkar et al., 2024).

Steering Vector Configuration We apply activation steering simultaneously to three different layer groups, targeting different representation stages:

- **Early semantic:** lower layers where token-level semantics develop (Geva et al., 2023);
- **Broad mid:** middle layers where semantic integration and emerging reasoning form (Jin et al., 2025);
- **Late reasoning:** upper layers where higher-order reasoning and task-specific representations are shaped (Song et al., 2026).

Since total layer counts differ across models (e.g., 16 for Llama-3.2-1B-Instruct vs. 36 for Qwen-2.5-3B-Instruct), the exact layer indices for each group vary. We report all ranges in Table 5. Unless otherwise noted, we sweep the steering coefficient $\alpha \in \{0.0, 0.1, \dots, 1.5\}$. We deliberately cap this range at 1.5, as preliminary extended ablations up to $\alpha = 7.5$ revealed severe performance degradation at higher steering strengths (Figure 5a). Specifically, we observed that excessive coefficients cause catastrophic representation collapse in Llama models and highly erratic instability in Qwen models. We provide a detailed discussion of this out-of-distribution behaviour in Section D.

Languages These experiments were conducted across 8 diverse languages, intentionally selected to span multiple dimensions of evaluation. These languages (French, Russian, Chinese, Bengali, Swahili, Yoruba, Wolof, Zulu) encompass variances in both resource representation and linguistic typology. This ensures a broad distribution of baseline model mastery, enabling the evaluation of steering techniques across varying degrees of reasoning capability and demonstrating that our method is largely agnostic to specific language families or resource constraints.

Evaluation Metrics The primary metric for all experiments is zero-shot accuracy. Since MMLU-ProX and our transfer benchmarks are multiple-choice tasks, we evaluate accuracy by parsing the model’s generated text for the correct option letter (e.g., A, B, C, or D).

Experiment	# Pairs	Setup
Baseline 1	200	Random pairs; diagnostic evaluation
Baseline 2	–	Gaussian noise; diagnostic evaluation
Baseline 3	20	Law $\rightarrow$ Math; diagnostic evaluation
Baseline 4	20	MMLU-ProX $\rightarrow$ Global-MMLU
Baseline 5	20	MMLU-ProX $\rightarrow$ Belebele
Multi-cat. (small)	20	Per category; diagnostic evaluation
Multi-cat. (full)	100	Per category; all held-out evaluation
CAST	50+2 $\times$ 50	Per category; all held-out evaluation

Table 1: Summary of experiments. *Diagnostic*: held-out questions correct in EN but incorrect in the target language. *All held-out*: all remaining target-language questions. CAST uses one behaviour vector (50 question pairs) and two condition vectors (50 question pairs each).

4.1 List of Experiments

Unless otherwise noted, all steering vectors are constructed from contrastive pairs as described in Section 3.3, where the base model succeeds in English but fails in the target non-English language.

Table 1 summarises all experiments. Baselines 1 and 2 verify that both learned directions and pairs filtered for correctness are necessary. Baseline 3 tests cross-category transfer (law $\rightarrow$ math), while Baselines 4 and 5 test cross-dataset transfer to Global-MMLU and Belebele respectively. The multi-category experiments first confirm per-category efficacy on a diagnostic subset, then scale to all held-out questions. CAST experiments add two condition vectors to mitigate the negative answer flipping observed under unconditional steering.

5 Results and Discussion

We focus on **Yoruba** as the illustrative target language. Baseline accuracy is lowest, thus the gap from English on MMLU-ProX is largest (Table 4). This makes Yoruba a strict test of whether English-derived steering can help when the model is already weak in the target language (L2). Yoruba is also tonally and typologically distant from En-

glish. These properties reduce the risk that improvements are driven mainly by superficial overlap with languages closer to English in pretraining data and lexicon.

5.1 Baselines 1 and 2

We find that injecting a random Gaussian noise vector in place of a learned steering vector yields erratic and generally detrimental results across all models and layer groups (Figure 5b), confirming that meaningful improvements require a purposefully extracted direction in activation space. Similarly, steering with vectors derived from randomly selected English versus Yoruba prompt pairs without filtering for correctness results in noisy accuracy trajectories which fail to produce consistent improvements (Figure 5c).

5.2 Baselines 3, 4, and 5

Baselines 3 through 5 collectively examine the extent to which the extracted steering directions generalise beyond their training setting. We observe improvements across all models, which may indicate that the extracted direction encodes a general cross-lingual signal rather than a purely domain-specific one (Figure 5d). Additionally, the steering vectors extracted from MMLU-ProX produce accuracy improvements when applied to questions in Global-MMLU and Belebele (Figures 5e and 5f), providing further evidence that the intervention is domain and task agnostic. Taken together, these transfer results suggest that the steering vectors capture information at the level of cross-lingual representation rather than task-specific content.

5.3 Multi-Category Experiments

Under the diagnostic evaluation restricted to questions answered correctly in English but incorrectly in the target language, we consistently observe accuracy improvements across languages and models (Figure 3a). Low-resource languages benefit more than high-resource ones, consistent with the hypothesis that weaker non-English representations have more room for cross-lingual transfer.

However, when extending to all held-out questions in a given category, performance drops across the board (Figure 3b). The accuracy improvement is noisy and inconsistent across steering coefficients (Figure 4a). Indiscriminate steering can harm questions the model would have answered correctly, as the direction may push activations away from already-correct representations.

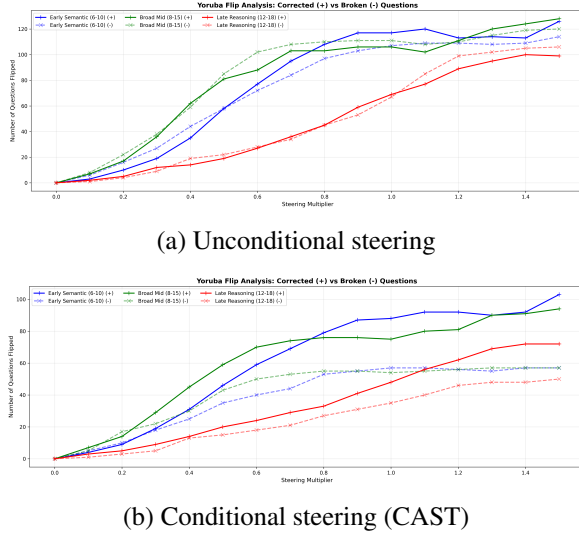


Figure 2: Flip analysis for Qwen-2.5-0.5B (Yoruba Chemistry). (a) shows convergence between corrected (solid) and broken (dashed) questions. (b) demonstrates a clear, expanding gap where corrections consistently outpace regressions.

This phenomenon is visually evident in our flip analysis (Figure 2a), which shows unconditional steering breaking nearly as many existing correct answers as it fixes.

5.4 Conditional Activation Steering (CAST)

With CAST, the improvement becomes substantially more stable, with accuracy gains observed over a broader range of α values and layer groups (Figure 4b). This stability is driven by CAST’s ability to selectively correct failures without degrading existing performance, creating a clear gap between corrected and broken questions (Figure 2b). Whereas the original work in (Lee et al., 2025) applied CAST to refusal behaviour, this extends the applicability of CAST to cross-lingual reasoning, where the condition detects whether the model is struggling with a non-English input.

Table 2 compares unconditional (SV) and conditional (CSV) steering across all models and languages for the law domain. CSV consistently outperforms SV, often by a wide margin.

To quantify the overall effectiveness of our approach, we aggregate results across all full multi-category sweeps and CAST runs (Table 3), explicitly excluding baselines, cross-dataset transfers, and diagnostic subsets. Here, a single configuration (*setting*) represents a full sweep over steering strengths $\alpha \in [0, 1.5]$ for a specific model, target

language, layer group, and subject category.

Under this rigorous evaluation, CAST yields positive target-language gains in 85.9 % of configurations versus 50.0 % for unconditional steering, and outperforms it in 81.3 % of paired comparisons. Low-resource languages benefit more (mean 2.03 pp) than high-resource ones (1.27 pp), which is consistent with greater headroom for cross-lingual transfer.

5.5 Mechanisms of Cross-Lingual Steering

Layer dynamics reveal that while interventions in early semantic and broad-mid layers typically recorded greater magnitudes of improvement, these gains were often accompanied by performance spikes and lower stability. In contrast, late reasoning layers yielded the most stable improvements, aligning with findings that complex reasoning is robustly processed within deeper layers (Jin et al., 2025).

Cross-lingual routing likely explains these results, as multilingual models appear to route non-English inputs through computation connected to English-internal processing (Schut et al., 2025). Consequently, contrastive pairs isolating "English-correct but L2-wrong" behaviour align with a direction that moves representations toward successful English reasoning on the same underlying task, explaining why random or unfiltered directions behave erratically (Section 5.1).

Steering strength (α) exhibits a non-monotonic relationship with target-language accuracy: small α leaves the model near its default computation, moderate α yields the largest gains, and large α can degrade accuracy by pushing activations outside the region of valid pretrained dynamics.

Latent capability ceilings remain despite steering successfully narrowing the English–non-English gap, as absolute MMLU-ProX scores are still modest for the smallest models (Llama-3.2-1B, Qwen-2.5-0.5B). As noted by Maraia et al. (2026) and Pokharel et al. (2026), steering amplifies existing latent reasoning but cannot synthesise it if the base model’s cross-lingual structure is weak, indicating this method complements rather than replaces continued multilingual pretraining (Kaplan et al., 2020).

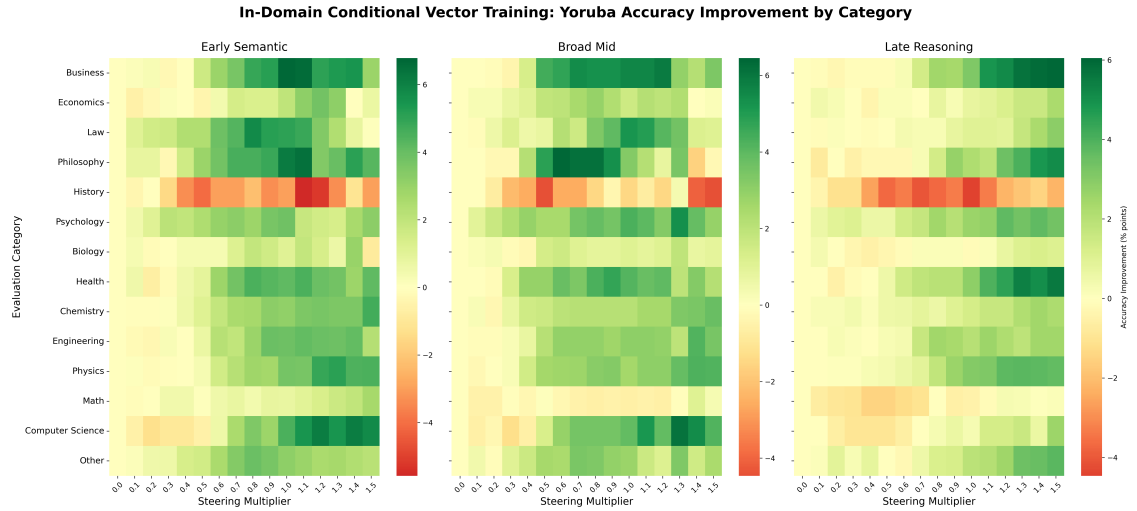
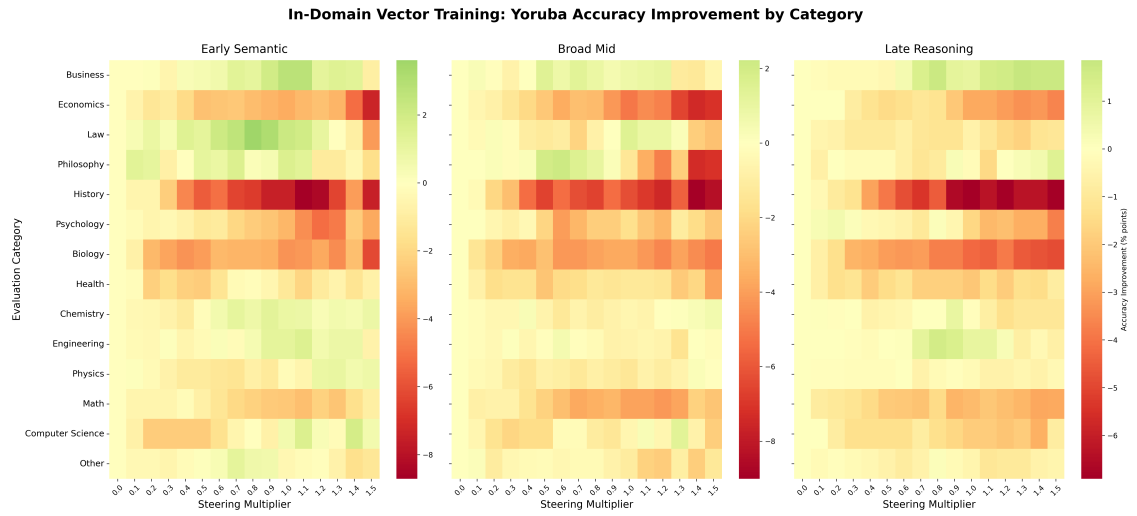
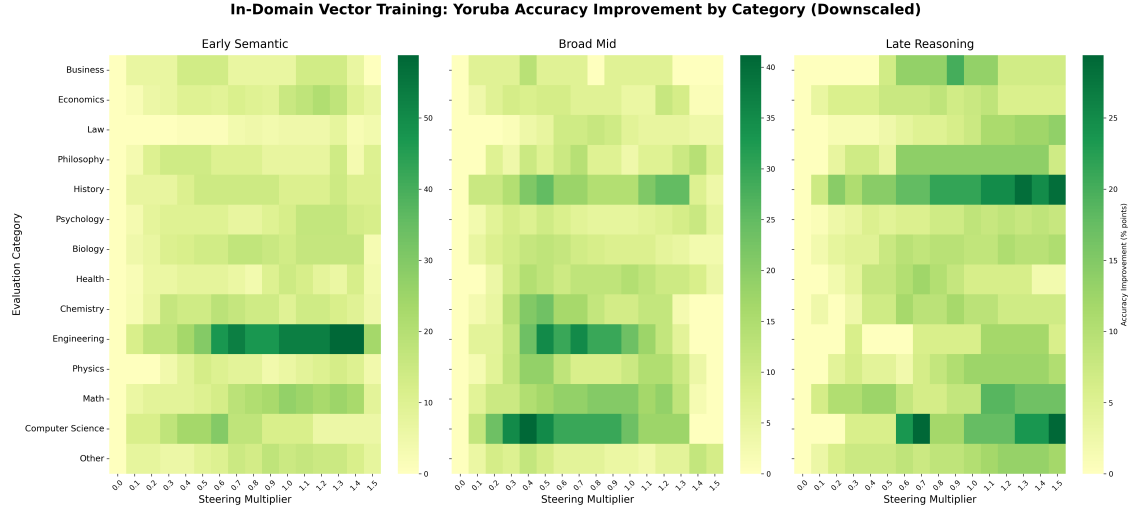


Figure 3: Category-level accuracy change (Δ) in raw percentage points for Qwen-2.5-0.5B on Yoruba across all 14 MMLU-ProX categories. (a) Diagnostic. (b) Full sweep. (c) CAST.

	BN		FR		RU		SW	
	SV	CSV	SV	CSV	SV	CSV	SV	CSV
LLaMA 1B	-0.698	-0.708	-0.231	5.56	0.231	2.97	0.815	0.848
LLaMA 3B	0.349	3.14	-0.233	0.989	0.582	0.865	-0.116	4.25
Qwen 0.5B	0.815	4.55	0.896	7.79	1.33	7.05	-0.116	2.29
Qwen 1.5B	1.86	3.21	0	0.989	0.931	1.48	0.582	1.36
Qwen 3B	0.466	1.11	0.232	1.48	0	1.36	0.349	1.95
	WO		YO		ZH		ZU	
	SV	CSV	SV	CSV	SV	CSV	SV	CSV
LLaMA 1B	1.05	1.78	3.26	4.48	0.116	3.17	2.21	3.19
LLaMA 3B	1.86	2.05	0.815	3.33	0.116	0.494	0.582	3.94
Qwen 0.5B	1.96	5.49	3.61	5.78	1.65	4.33	0.925	4.5
Qwen 1.5B	0.233	1.23	1.05	0.989	0.233	1.11	0.931	1.85
Qwen 3B	1.4	1.23	0.931	2.44	-0.349	1.24	2.79	3.41

Table 2: Comparison of model performance **improvement** (in raw percentage points) with and without conditional steering, for the domain of law. The subcolumns refer to steering without (SV) and with conditional steering (CSV). It is worth noting that the baselines differ between subcolumns, because in CSV more questions are taken out to train the conditional vectors.

Table 3: Aggregate steering results across full multi-category sweeps. **Positive** indicates the percentage of settings where the optimal α improves target-language accuracy. **High-resource** languages: FR, RU, ZH; **Low-resource** languages: BN, SW, WO, YO, ZU.

	Settings	Positive (%)	Mean Δ (pp)
Steering method			
SV	1,845	50.0	+0.42
CSV	1,722	85.9	+1.76
Paired comparison			
CSV > SV	81.3 % of 1,722 pairs		—
CAST by resource level			
High-resource	630	—	+1.27
Low-resource	1,092	—	+2.03

6 Conclusion

We investigated cross-lingual activation steering as a gradient-free method for narrowing the multilingual reasoning gap in small language models with at most 3B parameters. Using contrastive pairs drawn from MMLU-ProX where the model succeeds in English but fails in the target language, we extracted steering vectors via PCA-based aggregation (Lee et al., 2025) and evaluated both unconditional steering and CAST across 2 model families and 8 typologically diverse languages.

On diagnostic subsets restricted to questions the model fails in the target language, steering consistently improves accuracy, while baselines using random Gaussian vectors or randomly paired

prompts confirm that both learned directions and pair quality are necessary. When evaluation is extended to all held-out questions in a category, unconditional steering can flip previously correct answers, but CAST can mitigate this. Transfer experiments across categories and benchmarks further indicate that the learned directions encode cross-lingual rather than task-specific information.

6.1 Limitations

While our results demonstrate that CAST effectively mitigates cross-lingual reasoning gaps, we note the following boundaries of our current study:

- **Task Format:** Relying exclusively on multiple-choice benchmarks, transferability to open-ended generation remains untested.
- **Linguistic Diversity:** While covering eight diverse languages, the omission of further tonal languages beyond Chinese and Yoruba leaves open questions regarding typological interactions.
- **Data Budgets:** Our contrastive-pair budgets are modest (20–200 pairs/configuration), resulting in small diagnostic subsets.

Future Work Building on these findings, primary directions for future work include incorporating input-dependent, automatic α selection, scaling analysis to models beyond 3B parameters and conducting a systematic study of the trade-off between English degradation and target-language improvement across all configurations.

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Table 4: MMLU-ProX accuracy(%) across languages and instruct models without steering. Across a single model, the **bolded** number is the strongest performer, while the underlined number is the 2nd strongest.

Model	EN	BN	FR	RU	SW	WO	YO	ZH	ZU
Llama-3.2-1B	18.54	14.07	17.10	17.05	14.89	13.91	12.92	<u>17.52</u>	12.98
Llama-3.2-3B	30.25	17.60	<u>24.66</u>	24.80	19.00	17.45	15.43	24.43	16.33
Qwen-2.5-0.5B	<u>14.63</u>	11.55	13.26	13.35	11.75	11.77	11.53	15.49	11.64
Qwen-2.5-1.5B	27.53	17.28	23.40	22.26	16.97	17.29	16.15	<u>25.18</u>	16.70
Qwen-2.5-3B	35.38	21.71	30.59	29.77	20.91	21.45	20.68	<u>32.22</u>	20.89

A Benchmarking

Table 4 reports the baseline accuracy of all models across 9 languages without any activation steering. Across all models, English consistently achieves the highest accuracy, confirming the expected performance gap discussed in Section 1 and corroborating the findings of (Li et al., 2024), who quantified a systematic performance disparity between high- and low-resource languages across a range of language models. Chinese emerges as the second-strongest language for most models, which is consistent with its relatively strong representation in the pretraining corpora of both Llama and Qwen (Qwen et al., 2025). Low-resource languages such as Yoruba, Wolof, and Zulu consistently exhibit the weakest performance, with accuracy differences of up to 15 percentage points compared to English in the case of Qwen-2.5-3B. This gap is most pronounced in the smallest models (Llama-3.2-1B and Qwen-2.5-0.5B), where non-English accuracy frequently approaches chance level; this is an observation consistent with the scaling laws described by (Kaplan et al., 2020), which predict that smaller models allocate less capacity to underrepresented features of the training distribution.

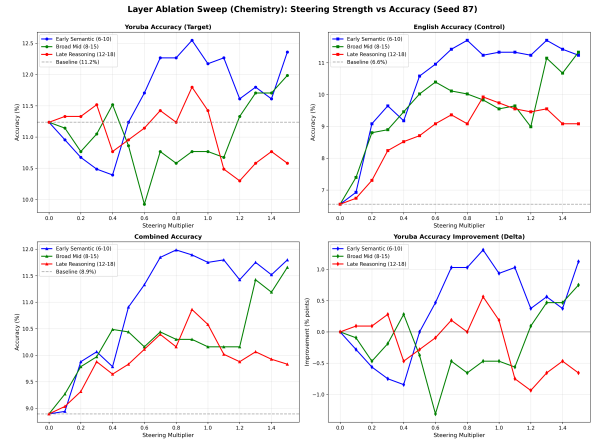
B Layer Grouping

Table 5: Layer indices per model and configuration.

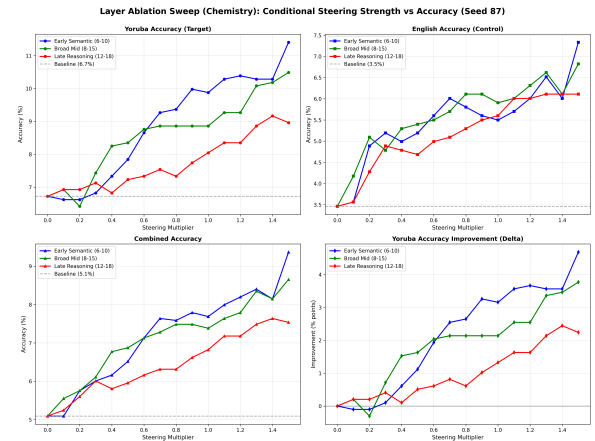
Model	Early	Broad mid	Late
Llama-3.2-1B	4–7	5–10	8–12
Llama-3.2-3B	8–13	10–18	15–19
Qwen-2.5-0.5B	6–10	8–15	12–18
Qwen-2.5-1.5B	7–12	9–17	14–21
Qwen-2.5-3B	9–16	11–22	18–27

C Supplementary Results

This section provides the full numerical results and accuracy evaluations referenced in the main text.



(a) Unconditional steering



(b) Conditional steering (CAST)

Figure 4: Accuracy trajectories for Qwen-2.5-0.5B (Yoruba Chemistry). (a) is highly sensitive to α and lacks a stable plateau. (b) yields improvements that are substantially more stable across α and layer groups.

	BN	FR	RU	SW	WO	YO	ZH	ZU
LLaMA 1B	27.5	24.2	21.7	25.7	19.2	21.4	26.4	24.3
LLaMA 3B	23.0	25.0	16.0	19.0	20.0	17.0	22.0	20.0
Qwen 0.5B	34.7	36.8	27.3	14.5	19.6	27.1	33.3	40.8
Qwen 1.5B	23.7	27.7	20.0	18.0	17.0	17.0	28.8	20.0
Qwen 3B	19.0	30.3	19.5	18.0	21.0	19.0	29.1	18.0

Table 6: Best percentage improvement (over a baseline of 0) on a small set of questions that each model gets wrong, when applying steering. α values swept from $[0, 7.5]$ in 0.5 increments. Note the maximum tends to be achieved for $\alpha \leq 2$.

	BN		FR		RU		SW	
	B1	B2	B1	B2	B1	B2	B1	B2
LLaMA 1B	0.5	1	1	2.5	1	3.5	1	2.5
LLaMA 3B	3.5	6.5	0	1	0	1	-0.5	0.5
Qwen 0.5B	3	1	1.5	1.5	2.5	1.5	3	2.5
Qwen 1.5B	1.5	2	1	1	2.5	2.5	1	2
Qwen 3B	0.5	1.5	2	0.5	3.5	2	2.5	0.5

	WO		YO		ZH		ZU	
	B1	B2	B1	B2	B1	B2	B1	B2
LLaMA 1B	2.5	2	2	1	1	1	1	1
LLaMA 3B	3	1	1.5	1	8.5	3.5	3.5	1
Qwen 0.5B	1.5	3	2.5	2.5	3	1.5	2.5	2.5
Qwen 1.5B	1	0.5	1	0	4	2.5	1.5	1
Qwen 3B	1.5	1.5	0.5	1.5	2	2.5	1.5	1

Table 7: Best percentage improvement (across $\alpha \in [0, 1.6]$) over Baselines 1 and 2. (Both experiments share the same baseline from which the improvement is calculated)

D Out-of-Distribution Activation Perturbation at High Steering Strengths

As observed in our preliminary extended sweeps (Figure 5a), applying excessively large steering coefficients (e.g., $\alpha > 2.0$) results in severe performance degradation rather than a graceful decline in reasoning capability. We attribute this to out-of-distribution (OOD) activation perturbation.

Activation steering functions via an additive intervention ($h' = h + \alpha v$), applying a small to moderate α successfully shifts the activation state within or very close to the model’s pretrained activation manifold. Subsequent layers remain well-calibrated to process these representations. However, massive multipliers aggressively push the hidden states entirely outside the geometric boundaries the model explored during pretraining.

This OOD perturbation manifests differently depending on the architecture:

- **Catastrophic Collapse (Llama):** For Llama models, extreme values push representations so far out of distribution that subsequent layers fail to process them meaningfully (e.g.,

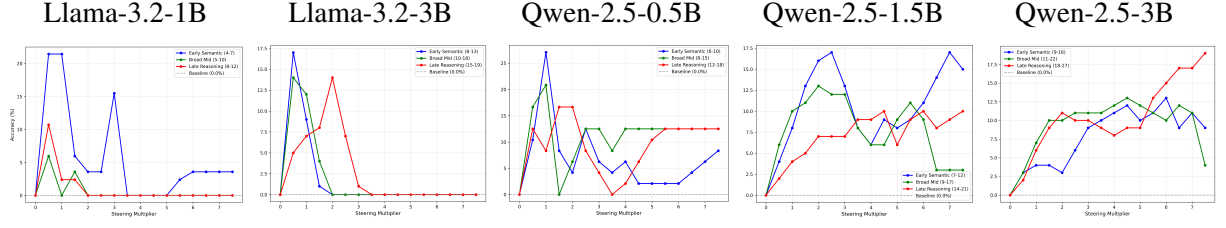
widespread activation saturation), effectively breaking the forward pass and dropping accuracy to near zero.

- **Erratic Instability (Qwen):** For Qwen models, the activations are pushed into untrained, noisy regions of the latent space. Subsequent layers multiply these unfamiliar activations by their weight matrices, resulting in chaotic, high-variance outputs that swing unpredictably across the sweep.

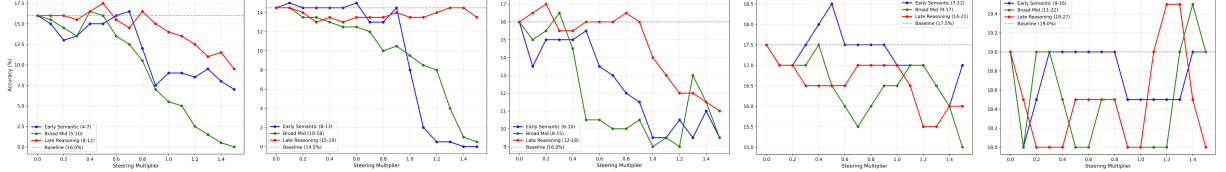
Consequently, we restrict our primary experimental evaluations to $\alpha \in [0, 1.5]$, ensuring the interventions remain within a structurally valid region of the models’ latent space.

E Per-item accuracy under CAST

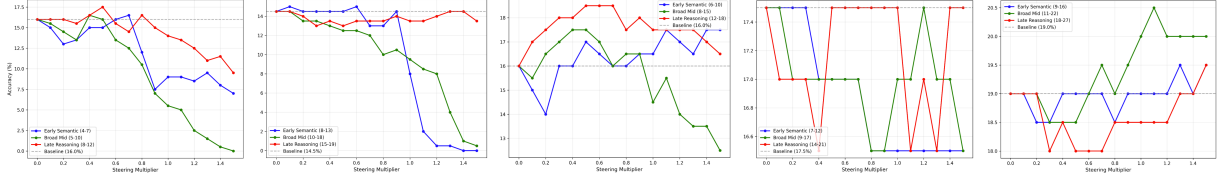
To assess the effect of steering at the item level, we compare each test item’s correctness before and after applying CAST at the optimal steering strength α^* . A *positive flip* denotes an item answered incorrectly at baseline but correctly after steering; a *negative flip* denotes the converse. The net flip count (positive minus negative) quantifies



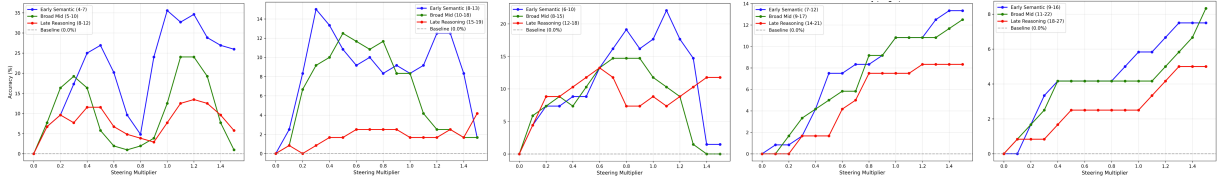
(a) Preliminary: Extended sweeps ($\alpha \in [0, 7.5]$) demonstrate that excessive steering pushes activations out-of-distribution, causing catastrophic collapse in Llama models and erratic instability in Qwen models.



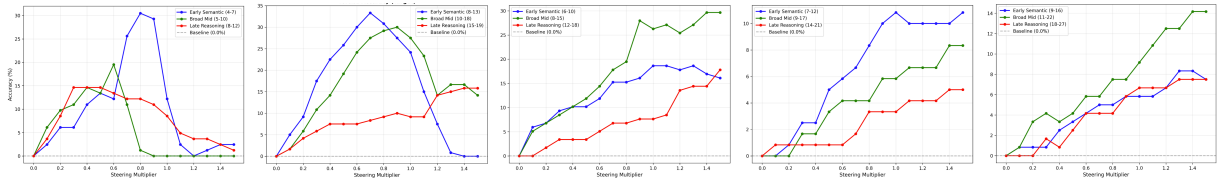
(b) Baseline 1: Erratic trajectories confirm that a learned steering direction is necessary.



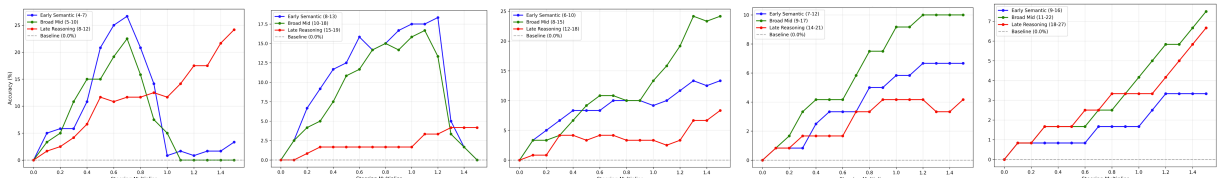
(c) Baseline 2: Noisy accuracy trajectories indicate that pair quality is essential.



(d) Baseline 3: Accuracy sweeps across layer groups and steering strength α .



(e) Baseline 4: Evaluating cross-dataset transfer within the same subject area.



(f) Baseline 5: Evaluating cross-dataset transfer across randomly selected, unaligned subject areas.

Figure 5: Accuracy trajectories across all baseline experiments (a–e) and model architectures on Yoruba. The x-axis of each plot represents steering strength α , and the y-axis represents task accuracy.

the number of items that CAST recovers beyond those it disrupts.

Across all CAST configurations, the mean net flip count per sweep is +11.4 (Qwen-2.5) and +11.6 (Llama), indicating that CAST consistently recovers more items than it impairs (Table 8). For EN-Yoruba, the mean increases to +14.5. Notably, steering at α^* preserves or marginally improves English accuracy, with fewer than 16 % of configurations exhibiting any English degradation. Figure 6 plots each sweep as a point in the target-language vs. English accuracy plane.

Table 8: CAST at optimal α^* : English accuracy change and net target-language flips (positive minus negative). The EN-YO row pools both model families.

Family	English change (pp)	English drop (%)	net flips
Qwen-2.5	+2.0	12.6	+11.4
Llama	+2.0	13.8	+11.6
EN-YO (both)	+1.5	15.3	+14.5

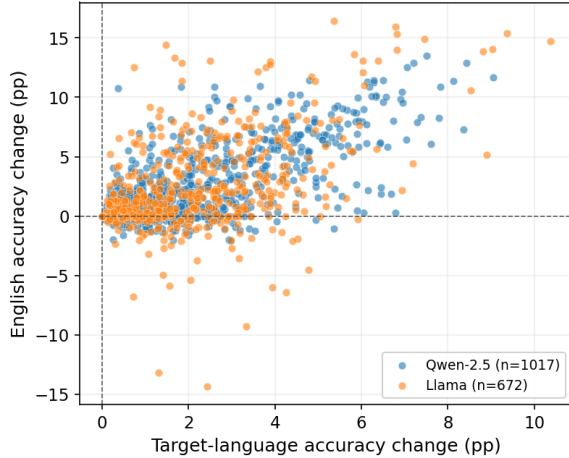


Figure 6: CAST at optimal α^* : target-language accuracy change vs. English accuracy change across all conditional steering configurations. Each point represents one sweep. Qwen-2.5 in blue, Llama in orange.

Side effects on English accuracy. Steering toward a target language could degrade English performance. To quantify this, we measure the change in English accuracy between the steered model at α^* and the unsteered baseline.

In practice, English accuracy is largely preserved or marginally improved: the mean English change is +2.0 pp for both Qwen-2.5 and Llama and fewer than 14 % of configurations exhibit any English degradation (Table 8). For the EN-Yoruba subset, the mean change is +1.5 pp with 15.3 % of configurations negative. Figure 6 shows the same relationship for every sweep: substantial target-language improvements rarely coincide with large English losses.

Case study: MMLU Law on Yoruba. To illustrate the interplay between target-language gains and English side effects within a single MMLU category, we examine *Law* for the EN-Yoruba pair across all layer groups and model checkpoints (15 conditional CAST sweeps).

We apply two fixed thresholds: a sweep counts as a *strong target gain* if the target-language improvement is at least 4 pp, and as a *strong English degradation* if English accuracy drops by 5 pp or more. Of the 15 sweeps, 4 achieve a strong target gain, 2 exhibit strong English degradation, and none satisfies both criteria simultaneously, indicating that substantial Yoruba improvements do not come at the expense of English.

The two cases of strong English degradation both involve Llama-3.2-3B: the Early Semantic group (−9.3 pp English, +3.3 pp Yoruba) and Broad Mid (−6.8 pp English, +0.7 pp Yoruba). In contrast, several smaller checkpoints achieve target gains of 4 pp or more with neutral or positive English accuracy, notably Qwen-2.5-0.5B Early Semantic (+5.8 pp Yoruba, +5.4 pp English) and Llama-3.2-1B Early Semantic (+4.5 pp Yoruba, +1.1 pp English). Table 9 reports all 15 sweeps with their per-item flip counts.

Why larger models are more susceptible.

We hypothesise that this pattern reflects how language-specific circuits are organised across model scales. In larger models, English reasoning is encoded in more specialised, localised pathways; injecting a steering vector into early or mid layers can therefore selectively disrupt English-specific representations without equally benefiting the target language. In smaller models, linguistic and reasoning representations are shared more broadly across languages due to limited capacity. The same intervention thus shifts both languages in the same direction, making it harder to harm English while improving Yoruba.

Table 9: MMLU *Law* (EN-Yoruba), conditional CAST at optimal α^* . **tgt** and **Eng.** report the accuracy change in percentage points for Yoruba and English, respectively. **net** is the number of items corrected minus items broken by steering on the target language. Runs marked $\uparrow$ achieve a target gain of at least 4 pp; runs marked $\downarrow$ exhibit an English drop of 5 pp or more. No run triggers both criteria.

Model	Layers	α^*	tgt	Eng.	net
Qwen-2.5-0.5B					
	Early Sem.	0.8	+5.8 $\uparrow$	+5.4	+48
	Broad Mid	1.0	+5.2 $\uparrow$	+6.9	+43
	Late Reas.	1.5	+2.9	+6.3	+24
Qwen-2.5-1.5B					
	Early Sem.	0.5	+0.7	0.0	+6
	Broad Mid	1.3	+1.0	0.0	+8
	Late Reas.	0.7	+0.4	0.0	+3
Qwen-2.5-3B					
	Early Sem.	0.7	+0.9	+0.9	+7
	Broad Mid	1.5	+0.7	+1.5	+6
	Late Reas.	1.5	+2.4	+1.1	+20
Llama-3.2-1B					
	Early Sem.	0.8	+4.5 $\uparrow$	+1.1	+38
	Broad Mid	0.4	+4.2 $\uparrow$	+2.5	+36
	Late Reas.	0.8	+3.3	+3.3	+28
Llama-3.2-3B					
	Early Sem.	1.2	+3.3	-9.3 $\downarrow$	+28
	Broad Mid	0.9	+0.7	-6.8 $\downarrow$	+6
	Late Reas.	1.2	+1.7	-0.6	+14

Additionally, the Broad Mid configuration spans the widest layer range and overlaps with both semantic and reasoning layers. For Llama-3.2-3B, this aggressive coverage appears to perturb the mid-layer representations that English relies on most, explaining the strong degradation (-6.8 pp English) despite only a modest Yoruba gain (+0.7 pp). By contrast, the more targeted Late Reasoning group intervenes after the critical English encoding stage and exhibits less English harm across all checkpoints.

Finally, Yoruba is an extremely low-resource language for these models: baseline accuracy is low, so there are many items that can be recovered by steering, while English accuracy starts higher and is therefore more fragile. This asymmetry amplifies the apparent trade-off in larger models where English circuits are more precisely tuned and thus more sensitive to perturbation.